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GENDER AND WORK PATTERNS IN INDIAN CITIES: A SOCIO-SPATIAL ANALYSIS

Vamsi Vakulabharanam and Sripad Motiram

ABSTRACT

Using an original household survey conducted in Hyderabad and Mumbai that identifies intra-city spatial coordinates of residents, this study presents a socio-spatial analysis of gender and paid work. The results show that the ease of movement through the city, allocation of care work-related considerations and educational attainment are all crucial to understanding the labor force participation patterns of urban women in India. A gender lens identifies key facets of access and mobility characterizing urbanization in developing countries. Spatial heterogeneity of residence has very different outcomes for the labor force participation of women and men.

KEYWORDS

Gender, space, inequality, labor force participation, Indian cities, segregation

JEL Codes: J16, J20, O18

HIGHLIGHTS

- Gender and city geographies mutually shape each other.
- Women's labor force participation varies across and within Indian cities; men's labor force participation is nearly uniform.
- Women's paid work is higher in cities that have women-friendly transportation.
- Women's paid work is higher in sub-city zones with superior transportation facilities for all.
- Policies that ease care or commuting burdens improve women's participation in the labor force.

Geography matters to the construction of gender, and the fact of geographical variation in gender relations, for instance, is a significant element in the production and reproduction of both imaginative geographies and uneven development.

–Doreen Massey (1994: 2)

INTRODUCTION

Cities in developing countries are driving global urbanization, but literature in economics has disproportionately focused on cities in the Global North.¹ Urban planning in developing countries reflects a gender-insensitive framework,² with adverse consequences (Ballal 2020). We address these gaps by focusing on gender relations in Indian cities. We make a case for exploring the mutual relationship between gender and space in the spirit of intersectionality (Collins 2019). Specifically, we hypothesize that cities function as gendered spaces, wherein gender and space mutually shape each other. We establish this hypothesis by focusing on the female labor force participation rate (FLFPR).³

Our study is the first to document variation in Indian intra-city FLFPR estimates.⁴ Focusing on the cities of Hyderabad and Mumbai, this article presents two findings. First, we find that the FLFPR varies across cities. Though not an unexpected result, we offer further explanation. Second, and more strikingly, we find that FLFPR varies quite significantly across different sub-city zones, while male LFPR is nearly uniform. We analyze this variation by investigating how gender interacts with space and other social cleavages (class, caste, and religion). In both Hyderabad and Mumbai, we show that whether a woman participates in the labor force or not is associated with a commuting burden, care burden, and educational attainment.

Gruesome incidents like the *Nirbhaya* case have brought to the forefront the lack of safe transportation for women in Indian cities, and challenges to urban mobility for women have been highlighted by academic studies.⁵ Yet, detailed location data needed for analysis are not provided in any databases (including the National Sample Survey [NSS], see the following for a detailed description).⁶ We therefore designed and conducted an original household survey in Hyderabad and Mumbai with detailed information on intra-city spatial units (both residence and work). We use a socio-spatial approach that conceives space as a product of socioeconomic and historical processes (see Chetty et al. [2014] for an early application in economics). We divide the cities of Hyderabad and Mumbai into zones based on their historical evolution. Using the above survey and drawing on the work of feminist geographers, we show how gender relations vary within and across Indian cities (Massey 1994).

There is considerable debate about FLFPR in India. Economists have largely focused on trends from the Employment and Unemployment surveys of the NSS. In rural India, a declining trend has been observed in recent decades. Some scholars (for example, Rangarajan, Iyer, and Seema [2011]) attribute the decline in rural FLFPR to development, namely, to increased enrollment in education and rising household incomes, but this explanation is contested (Kannan and Raveendran 2012). Indira Hirway (2012) argues for time-use data as a more accurate source for understanding women's work, given the nature of participation of Indian women (described as "sporadic" or "irregular").⁷ Studies on urban India document a low and stagnant trend and identify both demand-side and supply-side factors as culprits (Klasen and Pieters 2015). Some studies emphasize care obligations, arguing that women bear a disproportionate share of unpaid care work, which precludes labor force participation (Naidu 2016; Rao and Vakulabharanam 2018).⁸ We bring a fresh perspective to this debate by incorporating intra-city space. We have already presented our hypothesis about cities as gendered spaces. For our econometric analysis, our hypothesis is that whether a woman participates in the labor force or not depends upon individual (such as education), household (such as caste and care burden), and spatial characteristics (in particular, ease or difficulty of commuting to work). We establish this hypothesis using a probit analysis. As one would expect, high care and commuting burdens are negatively associated with a woman's ability to participate in the labor force. Our estimation has important policy implications – FLFP is positively associated with reliable and high-quality public transport and lower care burden of women.⁹ Overall, our analysis and findings indicate that space should receive far more consideration in the analysis of gender than it has received thus far.

DESCRIPTION OF DATA

We use data from a spatially representative survey that we conducted in the large cities of Hyderabad and Mumbai during 2015–17. Mumbai, a global mega-city is the capital of Maharashtra state (see Figure 1) and the financial capital of India. Hyderabad is the capital of Telangana state. The Indian Census uses the following spatial hierarchy: Urban Agglomeration (UA), District, Census Ward, and Enumeration Block (EB). Our survey covers Hyderabad and Mumbai UAs, by focusing on Hyderabad, Mumbai, and Mumbai Suburban districts.¹⁰ Our sampling design was multistage and stratified. Hyderabad district is divided into sixteen Subdistricts, which we used as strata and selected 1,000 households spread across 100 EBs (ten per each EB). Mumbai and Mumbai Suburban districts (together) are divided into twenty-four Municipal Wards. These are different from and larger than Census Wards, and we used them as strata and selected households as in

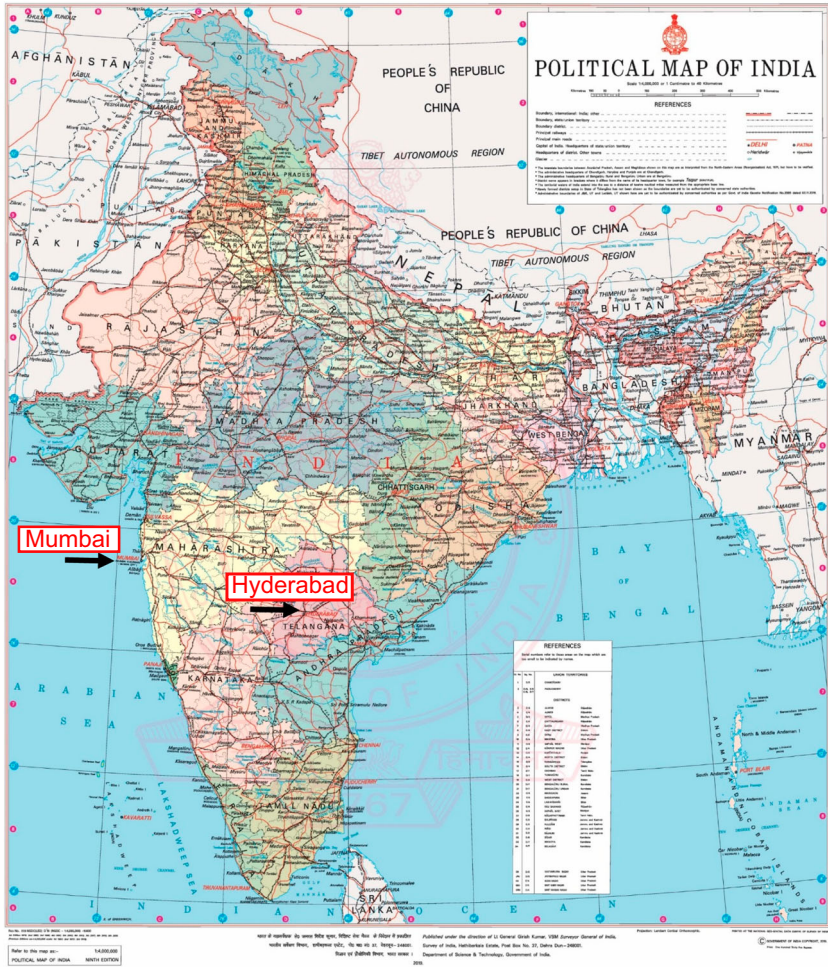


Figure 1 Location of Hyderabad and Mumbai.

Source: Survey of India, Government of India. <http://www.surveyofindia.gov.in/files/Political%20Map%20of%20India.jpg>.

Hyderabad (1,000 households in 100 EBs).¹¹ We administered a detailed schedule to selected households to collect information on several topics: demographic characteristics (for example, household size, caste, religion), education, income, consumption, migration status and details, details of work, and so on.¹²

To understand the interaction between gender and other cleavages, we divide the population into groups based on space, social group (caste

and religion), and class. Space is described in detail later on. Caste is an important social cleavage in India with Scheduled Tribes (STs), Scheduled Castes (SCs), and Other Backward Classes (OBCs) being the historically deprived groups.¹³ As is sometimes done, we group STs and SCs together as *Dalits*. In Indian cities, Muslims are another deprived group. Laurent Gayer and Christophe Jaffrelot (2011) and other scholars have highlighted this fact and also shown that the deprivation of Muslims is higher in urban areas (as compared to rural areas). Given this, we classify individuals into four *social groups*: Dalits (STs and SCs), OBCs, Muslims, and Others.¹⁴ In the regression analysis, we use a two-fold classification, combining the first three groups as *deprived*.

Another important social cleavage in India is class. There is a well-developed literature on countries in the Global North (such as in the US or UK) that divides the population into classes using frameworks influenced by Marx and Weber (Giddens [2009] presents an overview). A comparatively smaller literature exists on India (Patnaik 1987; Bardhan 1992; Vakulabharanam 2010; and Harriss-White 2018). The first two of these studies focus on rural/agrarian classes, whereas the third develops separate class schemas for rural and urban areas. Like in other countries in the Global South, the informal sector plays a substantial role in India, particularly as a source of employment (National Commission for Enterprises in the Unorganised Sector [NCEUS] 2008). Kalyan Sanyal (2009), in an influential study on the informal economy in developing countries, argues that the formal (“accumulation economy”) and informal (“need economy”) sectors are separate but also interdependent – the latter subsidizes the former (cheap labor) while also benefiting from it, through employment. We follow a broadly Marxian framework that builds on Vamsi Vakulabharanam (2010) and Sanyal (2009) to divide the population into the following six classes: three in the accumulation economy (Elite, Professionals, and Formal Workers); two in the need economy (Informal Owners and Informal and Self-Employed Workers); and Others (a residual category that includes retired workers). The National Classification of Occupations (NCO; Government of India 2004) provides a comprehensive list of occupations in India. Occupations are grouped into 1-digit divisions, which are further divided into 2-digit and 3-digit subdivisions. The 1-digit groups are: Legislators, Senior Officials, and Managers (1), Professionals (2), Technicians and Associate Professionals (3), Clerks (4), Service Workers and Shop and Market Sales Workers (5), Skilled Agricultural and Fishery Workers (6), Craft and Related Trade Workers (7), Plant and Machine Operators and Assemblers (8), Elementary Occupations (9), and Workers not classified by occupations (X).¹⁵ We used this classification and the information on occupation that we collected in our survey to group individuals into classes. For example, General Managers (3-digit code of 130) are classified under Elite and Architects, Engineers, and Related

Professionals (3-digit code of 214) are classified under Professionals. We also collected information on the status of the individual in the manner of surveys conducted by the NSS Office, for example, own-account worker in household enterprise, employer in household enterprise, regular salaried worker, casual worker, and so forth. We use this information to classify individuals as self-employed or informal workers.

Harriss-White (2018) highlights the importance of two classes in India: petty-commodity producers and merchants. She argues that these classes do not fit neatly into an orthodox Marxian two-class framework of capitalists and laborers and refers to these as “awkward classes.” Since we do not use a two-class scheme, we do not have a disagreement with Harriss-White’s analysis. Moreover, Informal workers and Self-employed in our scheme are broadly consistent with Harriss-White’s petty-commodity producers, and commercial capital (merchants) is also covered under Informal Owners. We adopt an analytically convenient approximation (O’Laughlin 1999) that makes a case for household-level investigations, by combining the public/domestic divide. In our analysis, for a household, we assign the highest class among its individual members to the household, for example, elite, if the household has a member in the elite class and another in the formal worker class. At the same time, we can identify paid (and unpaid) work patterns of different individuals in these households both inside and outside the domestic space.

In what follows, we present the descriptive statistics of some important variables and discuss gendered spaces.¹⁶

GENDERED SPACES OF INDIAN CITIES

We analyze gendered city spaces at two levels. First, we distinguish among different historically constituted spaces within a city from a gendered perspective and we examine gendered intra-city mobilities, that is, how women move from one space to another, public or private transport, and economic or cultural processes. Second, we compare across cities to show that the city-specific interaction of space and gender becomes a centrally defining feature of the city. Doreen Massey (1994) demonstrates the central role of region-specific gender relations in determining regional development in England. In the same spirit, we make comparisons across Hyderabad and Mumbai in the context of paid employment.

Gendered spaces within cities

Gendered city space in Hyderabad

Hyderabad is more than 400 years old and belongs to a class of Indian “walled cities,” in which a physical wall protected the urban

core.¹⁷ During the colonial period, India consisted of regions that were directly administered by the British and indirectly administered kingdoms. Hyderabad was the capital city of the largest of the latter group, the Nizam kingdom. A representative (administrator) of the colonial state was resident in the city.

Based on its historical evolution, we divide the city space of Hyderabad into four zones: Old-Walled city, Nizam's city, British-Resident city, and Neoliberal city. As its name suggests, the first zone (Old-Walled city) is the oldest region that was constructed by the Qutub Shahi dynasty in the sixteenth century. It is also the poorest (lowest average per-capita income in our survey). The second zone (Nizam's city) emerged during the eighteenth to nineteenth centuries when Hyderabad state was ruled by the Nizam, who broke away from Mughal rule. The third zone (British-Resident city) emerged during the nineteenth to twentieth centuries and includes the area where the British resident and army (Cantonment) were stationed. Finally, we term the area where the new economy emerged (a process that started in the 1970s) as the Neoliberal city. These zones and the subdistricts of Hyderabad are depicted in Figure 2.¹⁸ According to many social scientists, the economic reforms initiated in the early 1990s mark the beginning of the neoliberal phase in India (Steger and Roy 2010). As some scholars point out, neoliberalism has changed Indian cities, but its effects vary across areas within a particular city and also across different types of cities (for example, metros versus smaller cities; Banerjee-Guha 2009). The names that we have chosen for zones should be seen as descriptors and correspond to the character of the area that they represent and to the lived experience of the residents of the city. For example, the Old-Walled and British-Resident zones are what residents of Hyderabad refer to as "Old City" and "Secunderabad and Cantonment," respectively. The "Neoliberal" zone is where neoliberalism has left a distinctive imprint, through companies in newer sectors like finance and information technology (IT) and gated communities.

Given the role that commuting plays in our analysis, discussing transportation infrastructure and modes in Hyderabad is useful. We present a brief overview based on existing literature and our own knowledge of the city. In Hyderabad and other cities in Andhra Pradesh and Telangana, people use several modes of transport to commute to work. Some of the private modes are bicycle, scooter, motorcycle, car, jeep, and van.¹⁹ Paid auto-rickshaws and taxi services (like Ola and Uber) are also used. Public transport mainly consists of buses, and although there is a local/suburban railway system, it connects only a few areas. There has been an increase in reliance on private transportation in recent decades, largely because public transportation is unable to meet the needs of commuters (Vamshi Krishna and Chattaraj 2020). A major public

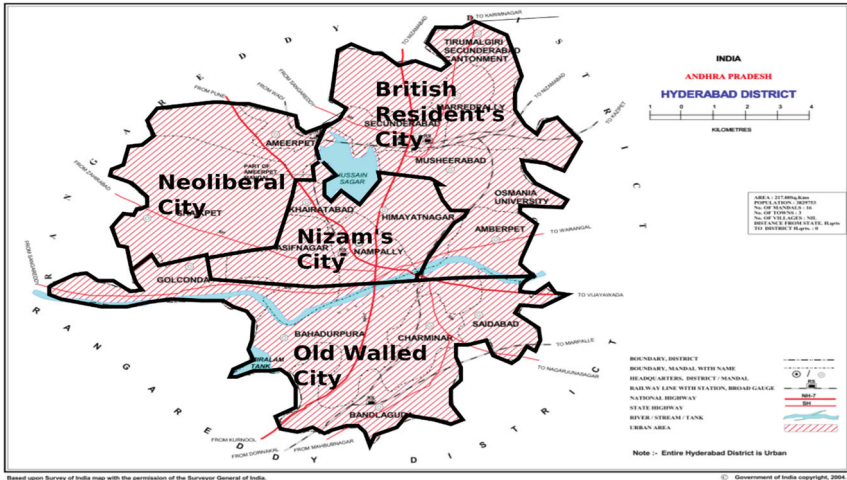


Figure 2 Map of Hyderabad district with sub-districts and zones

Source: District Census Handbook, Census of India 2011. We overlaid the zones on the map provided by the Census.

transportation initiative that has been taken up is the Hyderabad Metrorail. However, several experts are of the opinion that it has limited utility in serving common people and instead suggest augmenting the bus system (Ramachandraiah 2007).

Next, we draw on how axes like class, caste, and religion inscribe themselves on these historically constituted zones (Table 1). Our earlier work showed that the poorest Old-Walled city and the prosperous Neoliberal zone are the least diverse in terms of class, caste, or religion. The other two zones, Nizam's city and British-Resident city, have more mixed neighborhoods and deliver better education and lower poverty levels for their residents.

Finally, when we turn to FLFPR differences across city spaces, additional insights emerge. Table 1 shows the summary of paid employment for Hyderabad city, revealing that FLFPR (30.9 percent) is much lower than its male counterpart in our survey (> 80 percent). Another notable feature is that while male LFPR is almost uniform in various zones of the city, there are significant differences in FLFPR across zones. FLFPR is lowest in the Old-Walled city, highest in the Neoliberal city, and has intermediate values in the two zones that developed in the eighteenth and nineteenth centuries. The Old-Walled city zone (with a high proportion of Muslims) has been excluded from the economic dynamics of the developing city, turning it into a ghetto – Table 1 shows low representation of the elite and professional classes/occupations, and informal/self-employed worker

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Table 1 Space, social group, class, and gender (Hyderabad)

	<i>Old-Walled</i>	<i>Nizam's</i>	<i>British</i>	<i>Neoliberal</i>	<i>Hyderabad</i>
Social group^a					
Dalit (ST and SC)	12.12	12.67	20.56	16.01	14.58
OBC	27.92	49.81	60.08	43.46	42.37
Muslim	54.05	18.48	4.09	16.99	30.25
Others	5.92	19.05	15.27	23.53	12.80
	100	100	100	100	100
Class^b					
Elite	3.63	9.14	4.79	14.38	6.10
Professional	3.80	18.57	8.28	17.65	9.64
Worker (Formal)	28.42	22.76	34.53	21.90	27.98
Owner (Informal)	13.40	9.9	8.28	6.54	10.77
Worker (Informal) & self-employed	43.16	36.66	39.82	31.04	39.82
Others	7.59	2.95	4.29	8.50	5.69
	100	100	100	100	100
Gender					
College or higher ^c	9.54%	13.71%	17.33%	19.08%	13.23%
LFPR (Women) ^d	24.65%	37.88%	31.30%	38.94%	30.91%
NCO 2 or 3 Jobs ^e	14.26%	28.18%	18.04%	28.57%	20.30%

Notes: ^aPercentages of adults (older than 18 years, both men and women) who belong to various social groups. ^bPercentages of adults (older than 18 years, both men and women) who belong to various classes of households. ^cFigures for adult women. We included diploma after high school under college. ^dFigures for adult women. ^eOne-digit National Classification of Occupations (NCO). 2: Professionals; 3: Associate Professionals, and Technicians. Percentage of all (men and women) workers.

Source: Authors' computations from survey data.

households dominate. This zone also lags behind the rest of the city in the educational attainment of women, as well as employment that is somewhat women-oriented.²⁰ Commuting costs tend to be higher for women located in this part of the city given the limited nature of public transport and the range of available jobs. In contrast, the Neoliberal zone has much higher educational attainment for women and has nearby opportunities for women-oriented employment. There is a high proportion of elite, professional, and formal worker households. Even in the affluent Neoliberal city, there is evidence of dependent informality (Motiram and

Vakulabharanam 2020) – a high enough proportion of women come from informal labor or self-employed households. Public transport is not as prevalent in this zone as in the middling zones, but private transport such as household cars, Uber, and shared auto-rickshaws contribute to mobility and FLFPR.

The two middling zones are distinct, with Nizam's city having a higher FLFPR than British-Resident city. Nizam's city has emerged as the main employment hub of Hyderabad and continues to provide employment both to its own residents as well as to residents of other zones in the commercial and professional domains (including government employment). Well-developed public transport and the prevalence of commuting within this zone and between this and other zones results in a higher FLFPR than the British-Resident city. The British Resident city is also a major center of military establishments, which are mostly dominated by men.

The more positive results for mixed neighborhoods from our earlier analysis does not neatly correlate with FLFPR.²¹ This puzzle can be made sense of as follows. First, it is clear from the table that the higher the formal sector presence of women (elite, professional, and formal worker classes), higher the FLFPR (for example, Neoliberal zone). Second, heightened class inequalities play an important role in increasing FLFPR at the lower end of the occupational ladder (Arruzza, Bhattacharya, and Fraser 2019). Women working in the formal sector tend to typically transfer the care burden they face to informal sector women workers in their neighborhood, who are hired to perform that labor (Ehrenreich and Hochschild 2002; Benería and Floro 2006). The physical and social isolation of the Old-Walled city limits access to employment as paid care workers in professional households with working women. This lack of access increases the isolation and confinement of women in the walled city indicated by the much lower FLFPR for that zone. Low-paid informal work, especially paid care work, is more easily undertaken when proximity or convenient transport makes the work consistent with performing care work at one's own home. A gender and paid employment perspective, therefore, further complicates city space, bringing care work and mobility considerations into sharper focus. We elaborate on this later using regression analysis.

Gendered city space in Mumbai

Mumbai is a newer and larger city than Hyderabad. It emerged in the early colonial era and was directly administered by the British.²² As we discussed above, at present, Mumbai city is spread over two districts: Mumbai and Mumbai Suburban. The city emerged in the southern portion, which is in the former district. Population pressure, land markets, and state policies resulted in the spread of the city toward the north and into the suburban district. We divide the city into five zones: British and Neoliberal city,



Figure 3 Map of Mumbai with zones

Source: Municipal Corporation of Greater Mumbai (<http://dm.mcgm.gov.in/ward-maps>). We overlaid the zones on the map.

Old Industrial city 1, Old Industrial city 2, Western Neoliberal city, and Northern Neoliberal city. These zones are depicted in Figure 3.²³

First, we look at the historical processes that constituted these five zones. British city is the southernmost part of Mumbai, where British residents were located during the colonial rule along with various business establishments. We added “neoliberal” to this zone because during and after the 1990s, this is the space that became an important location for many of the multinational corporations that entered India in the wake of market-oriented reforms of the 1990s. Residents of Mumbai refer to this area as “South Mumbai” or “South Bombay.” Old Industrial City 1 and Industrial City 2 housed the factories and worker-residences that were created during the long waves of industrialization (mainly the cotton textile industry, but also for production of other commodities) that happened in Bombay during 1870–1940. The Western Neoliberal city (Bandra-Kurla area being an important part) is the city space that was restructured in

the 1990s to incorporate financial firms and headquarters of other kinds of multinational and national businesses.²⁴ The Northern Neoliberal city developed in the 1990s from a broadly residential zone to one including information technology, films, finance, and other new industries.²⁵

Second, in analyzing class, caste, and religion across city zones we found patterns similar to Hyderabad. In Mumbai too, the zones that have more mixed neighborhoods tend to have better development outcomes. In a different study, using an instrumental variable regression approach, we show that the patterns of older industrialization in Mumbai that created mixed residential neighborhoods in terms of class, caste, and religion have a positive impact on the development outcomes of an individual resident in various neighborhoods. The key point is that the more homogeneous neoliberal zones do not fare the best in terms of enhancing economic development for an average resident.

Looking at transportation in Mumbai, there are some differences from Hyderabad. Zeeshan Shaikh (2015) presents estimates from the latest Census. As in the case of Hyderabad, private modes of transport include bicycle, scooter, motorcycle, car, and so on, and people take recourse to paid auto-rickshaws and taxis. However, a very high number and percentage of population use public transport. Many commuters use buses and other modes of transport (for example, autorickshaws) to connect to the suburban train system. Given that Mumbai is a coastal city, water transport is also used by a small percentage of commuters. Problems of congestion (largely due to increasing private modes of transport) and inadequacy of existing modes of public transport have been highlighted by several studies and government reports (GOM 1987; Badami 2005; MCGM 2016).

Third, we analyze FLFPR in Mumbai in Table 2. While FLFPR is higher than in Hyderabad, it is lower (35.41 percent) than its male counterpart (> 85 percent). As in Hyderabad, male LFPR in different zones is nearly uniform across the city. FLFPR is relatively higher in three zones (> 35.41 percent): British, Western, and Northern Neoliberal cities, and it is lower in Old Industrial cities 1 and 2. The neoliberal zones have a relatively high proportion of educated women and concentration of households in the formal sector (elite, professional, and formal worker). In the three neoliberal zones that are huddled together in FLFPR terms, the percentage of women-oriented jobs is also relatively high. In Old Industrial cities 1 and 2, which house the poorer residents of Mumbai (for example, Industrial city 2 includes M ward, where Dalits were settled in a concentrated population), FLFPR is lower. In Industrial city 2, the educational attainment of women is also the lowest among all the five zones, and a high proportion of households are in the informal sector.

It is noteworthy that all of the relatively far-flung neoliberal city spaces in Mumbai exhibit relative homogeneity in terms of FLFPR. This is mainly

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Table 2 Space, social group, class, and gender (Mumbai)

	<i>B. Neo</i>	<i>O. Ind 1</i>	<i>W. Neo</i>	<i>N. Neo</i>	<i>O. Ind 2</i>	<i>Mumbai</i>
Social group^a						
Dalit (ST and SC)	19.04	24.15	16.35	19.49	18.62	19.14
OBC	13.96	19.65	9.70	8.15	20.54	12.86
Muslim	36.55	11.92	19.49	8.90	11.90	15.96
Others	30.46	44.28	54.46	63.45	48.94	52.04
	100	100	100	100	100	100
Class^b						
Elite	10.91	0.64	5.94	4.50	3.84	4.88
Professional	5.58	8.21	9.35	9.00	6.53	8.27
Worker (Formal)	23.86	26.41	26.31	22.12	19.00	23.86
Owner (Informal)	6.09	6.92	7.08	6.00	12.67	7.42
Worker (Informal) & self-employed	52.03	55.07	47.99	55.2	55.85	52.73
Others	1.52	2.74	3.32	3.19	2.11	2.83
	100	100	100	100	100	100
Gender						
College or higher ^c	21.31%	21.63%	24.32%	20.08%	14.58%	21.03%
LFPR (Women) ^d	38.89%	32.54%	38.64%	35.43%	28.90%	35.41%
NCO 2 or 3 Jobs ^e	11.56%	12.86%	15.90%	15.49%	12.00%	14.24%

Notes: ^aPercentages of adults (older than 18 years, both men and women) who belong to various social groups. ^b Percentages of adults (older than 18 years) who belong to various classes of households. ^cFigures for adult women. We included diploma after high school under college. ^dFigures for adult women. ^eOne-digit NCO for women in the labor force. 2: Professionals; 3: Associate Professionals and Technicians. Calculated over those who are working (that is, have an NCO code).

Source: Authors' computations from survey data.

because of the commuting distinctions between Hyderabad and Mumbai that are explained below. As in the case of Hyderabad, higher formal sector presence (elite, professional, and formal worker classes) of women in the neoliberal zones and the dependent informal paid employment that it generates is the main explanation for the higher FLFPR in these neoliberal zones, a complex intersection of class, gender, and space.

The differently gendered cities of Hyderabad and Mumbai

In popular lore, as well as in academic literature, Mumbai has been considered a safe city space for women (World Bank 2011). This has been attributed to a wider prevalence of safe public transport and a much longer history of urban experience that allows women to travel longer distances and late into the evening (Punwani 2013). Commuting patterns of men and women are similar in Mumbai (trains, buses, and private transport),

Table 3 Results of decomposition (Hyderabad)

	<i>Share</i>	<i>FLFPR</i>	<i>Contribution</i>
Social Group			
Dalit	15.35%	36.28%	18.02%
OBC	43.18%	37.02%	51.73%
Muslim	27.41%	20.57%	18.25%
Others	14.06%	26.40%	12.01%
	100%	30.91%	100%
Class			
Elite	7.49%	40.95%	9.92%
Professional	10.56%	31.08%	10.62%
Formal Worker	29.12%	33.82%	31.87%
Owner (Informal)	9.35%	35.11%	10.62%
Worker (Informal) & self-employed	37.12%	29.62%	35.57%
Others	6.35%	6.74%	1.39%
	100%	30.91%	100%

Notes: All computations for adult women (older than 18 years).

Source: Authors' computations from household survey data.

unlike in other Indian cities (Rukmini 2019). This does not mean that women do not suffer from indignities or harassment during travel, but in relative terms, Mumbai is better than most other Indian cities. In contrast, Hyderabad is not considered as safe and lacks decent public transport (Deshpande 2020). In fact, until 2016, local trains did not cover a wide network, so much of the commuting took place through buses or shared auto-rickshaws.

In terms of comparing levels of FLFPR in the two cities, our a priori expectation was that there would be a higher FLFPR in Mumbai as compared to Hyderabad. We see this in the survey data: 35.4 percent (Mumbai) as compared to 30.9 percent (Hyderabad). The greater safety of commuting in Mumbai and the higher demand and supply of women-oriented jobs, both at the higher and lower ends of the class ladder (Table 4), are the main reasons. Mumbai has a higher proportion of working women from the informal and self-employed classes (Table 4) compared to Hyderabad (Table 3). Many of these are slum dwellers, with Mumbai having a much higher proportion (41.3 percent) than Hyderabad (31.9 percent).²⁶

A second interesting comparison is the within-city differences in FLFPR. As compared to Mumbai, Hyderabad has larger variation of FLFPR for women across the zones. We attribute this to the fact that commuting is harder for women in Hyderabad, and they tend to work less if it entails a longer commute. In Mumbai, on the other hand, commuting in local-transit trains has a much longer history, and women have converted

Table 4 Results of decomposition (Mumbai)

	<i>Share</i>	<i>FLFPR</i>	<i>Contribution</i>
Social Group			
Dalit	19.01%	34.02%	18.27%
OBC	12.54%	30.19%	10.69%
Muslim	14.91%	42.32%	17.82%
Others	53.55%	35.19%	53.23%
	100%	35.41%	100%
Class			
Elite	5.6%	59.16%	9.36%
Professional	9.15%	35.34%	9.13%
Formal Worker	25%	38.17%	26.95%
Owner (Informal)	7.1%	27.77%	5.57%
Worker (Informal) & Self Employed	49.84%	33.70%	47.44%
Others	3.31%	16.67%	1.56%
	100%	35.41%	100%

Notes: All computations for adult women (older than 18 years).

Source: Authors' computations from household survey data.

necessary commuting into a complex socio-spatial phenomenon. Local trains in Mumbai have compartments/carriages that are exclusively for women. References in media and popular culture (and discussions with Mumbai women commuters) indicate that women in Mumbai have adapted to their commutes by transforming an inert movement between points A and B into a rich socio-cultural space. First, women use this space to construct social networks and as a pleasurable “women’s only space” where they can share consumption of various home-made items or compare fashion purchases.²⁷ Second, women tend to perform some of the care-related work (for example, cutting vegetables for evening meals) during their commutes.²⁸ Third, women also produce commodities that can be sold among themselves (such as embroideries or night gowns). Overall, commuting is not merely a necessary evil but becomes a space of new expectations, fulfillment, and even pecuniary benefits.

Tables 3 (Hyderabad) and 4 (Mumbai) report the FLFPR share of the population, and share of the female labor force for sub-groups (classes or social-groups).²⁹ A sub-group may be overrepresented (underrepresented) in the labor force relative to its population share. We focus on women belonging to different social groups and classes below.

In Hyderabad, for social groups, we observe that Dalit and OBC women are overrepresented in the labor force, whereas Others are underrepresented. The former (lower castes) are involved in less-remunerative occupations. Muslims are significantly underrepresented in

the labor force. When it comes to class, the notable observation concerns women from informal and self-employed worker households, who have a much lower LFPR than the ones in the formal sector.

In Mumbai, Dalits are slightly underrepresented, OBCs are slightly underrepresented, and Others are represented at their proportion. The truly notable result is that Muslims are over-represented unlike the case in Hyderabad. Coming to classes, women from elite and professional households tend to have higher LFPR. Women from informal and self-employed households have lower LFPR, although they participate much more than their counterparts in Hyderabad.

In both Hyderabad and Mumbai, gender interacts with class, caste, and religion in significant ways. Three points need emphasis in the overall comparison. Muslim women are under-represented in Hyderabad's labor force (not so in Mumbai). Muslim women's participation shows that mobility and access (more than religion or "cultural choice") determines FLFPR. The fact that the Old-Walled city in Hyderabad that has high Muslim concentration has been ghettoized while Muslims are not as ghettoized in Mumbai is a key point. Second, Dalits are overrepresented in Hyderabad but not in Mumbai, again reflecting ghettoization (in Mumbai's M Ward) as compared to Hyderabad, increasing the commuting burden for them. Third, Mumbai has overall a higher proportion of women from informal and self-employed families in paid employment in comparison to Hyderabad.

We find that gender is central to the experience and development of the cities of Hyderabad and Mumbai. Walled cities, ghettos, transportation infrastructure, and class inequalities weigh heavily in the determination of FLFPR.

ECONOMETRIC ANALYSIS OF FLFP

An empirical model

In the analysis above, we looked at how gender interacts with class, social group, and space. However, a multivariate analysis that simultaneously incorporates spatial and individual- and household-level variables can shed more light on participation of women in the labor force. At the individual and household levels, key variables are education, care burden, main occupation/source of income for the household, and the social group to which the household belongs. More education makes one eligible for better-paid jobs, potentially in the formal sector. One would also expect that better-educated women are more desirous of paid work. Women are, by norm in a patriarchal society, disproportionately obligated to undertake household care work. The need to do more care work for one's own household hinders the ability to participate in the labor force

(unless transportation or proximity enables participation in paid care work that is consistent with own-household care work). All else being equal, high income from other sources (for example, husband's employment) alleviates the necessity for paid work and may raise the reservation wage, making withdrawal from the labor force preferable to low-wage labor. Social norms and discrimination can have differential impacts on the ability/desire of women belonging to different social groups. Therefore, it is important to control for this household characteristic. The second set of factors is spatial.³⁰ If safe and adequate transport facilities are available in the neighborhood (zones), and commuting is not too costly in terms of time or money, then this would facilitate participation of women in the labor force as they can commute to work with ease. This would also increase women's ability to combine paid work with the performance of care work.

Given these two sets of factors, to shed further light on the determinants of FLFP, we estimate the following model:

$$FLFP_i = \alpha_1 + \beta_1 X_i + \gamma_1 S_i + u_i \quad (1)$$

i indexes individuals. $FLFP_i = 1$ if i participates in the labor force and 0 otherwise. X_i is a vector of characteristics of the individual i and the household to which she belongs. S_i describes the spatial unit that i resides in, and u_i is an idiosyncratic error term. As discussed above, under X_i , we include a variable that captures care burden and dummies for education, occupation of the household, and deprived status. We controlled for the occupation of a household by including a dummy for whether the household is an informal and self-employed one or not.³¹ We model the care burden with the ratio of the number of children and male adults in the household to the number of adult women, a version of the dependency ratio that captures both the need for care and the potential of shared care across other women in the household.³² Under S_i , we include the commuting ease/burden. Given the importance of zones, we proxy for commuting burden using the average distance to work in terms of zones traveled by men living in the neighborhood. We use the average for *men* to avoid the problem of endogeneity. We use Enumeration Block (EB) as the neighborhood because it is the lowest spatial unit and since commuting options (for example, public transport) are similar for residents of the same EB.³³

Two issues need to be discussed before we proceed with the estimation. First, what if households decide about the labor participation of women *and then* choose where to live (or make the labor participation and location decisions *together*)? This would result in a selection problem. Secondary literature suggests that this problem is not serious, and such households constitute a minor share of the population. The factors that are important for the location decision are: migration networks, safety concerns of the

community, discrimination in land and credit markets, real estate, and state policy. For example, migrants to cities rely on local kin or village networks to facilitate the process of migration, including job search, information, and stay in the city (Lall, Selod, and Shalizi 2006). There are several examples of areas that have formed in Indian cities through migration from a particular part of India, facilitated through networks (see for example, Katherine Boo [2014] for Mumbai). The most important motive for migration for women is marriage (Rao and Finoff 2015), whereas the economic motive (job) is important for men. Migrant women move to the place where their husbands are located. Unmarried women typically stay with their parents – in an area that has been decided by their parents (usually long ago and without consideration of work prospects of future generations).³⁴

Studies from several cities on Muslims point out that due to repeated bouts of “communal” violence (between Hindus and Muslims), Muslims have located themselves in certain parts of cities so that they have security in numbers (Gayer and Jaffrelot 2011; see Neena Ambre Rao and S. Abdul Thaha [2012] for Hyderabad). These studies also point out discrimination in land and credit markets against Muslims that restrict their location. State policy has also been responsible for locating certain communities in particular areas of cities; for example, M-ward, Mumbai (which is in the vicinity of a nuclear waste dump) was settled by bringing in Dalits and Muslims (Contractor 2012). We believe that the above literature strongly supports our regression. However, since we are not explicitly analyzing selection of individuals/households into spatial locations, our estimates can be interpreted as associations (not causal).

Second, transportation facilities and infrastructure can potentially emerge in response to the demands of women who desire to work. However, this is unlikely to happen in the short-term, which is the relevant period for individuals/households making decisions about work, or even in the medium term, as suggested by evidence since the 1990s (when India embarked on a neoliberal path). Several studies have suggested that infrastructure in Indian cities, particularly transportation infrastructure, is being dictated by considerations other than the needs and demands of the citizens. Instead, the state is catering to various lobbies, such as real estate and construction. There is also an attempt to showcase Indian cities as “world class” by building infrastructure that is not really necessary (a transition from an earlier focus on “livable cities”), which is consistent with what scholars have describes as “neoliberal urbanism” (Banerjee-Guha 2009). Mass Rapid Transport System (MRTS, more popularly Metro) projects have been initiated in several Indian cities (for example, Delhi, Mumbai, Hyderabad), and there are future plans for others, even small cities (such as Nagpur). Studies on the Delhi metro (Singh and Ravi 2018) have shown that it caters to a narrow section of the population

Table 5 Probit regression for female labor force participation (dependent variable: 1 if a woman participates and 0 if not)

	<i>Hyderabad</i>	<i>Mumbai</i>
Belongs to a deprived household	0.240** (0.115)	0.118 (0.078)
Higher Secondary or higher education	0.253* (0.083)	0.155** (0.080)
Belongs to an informal or self-employed household	0.011 (0.082)	−0.007 (0.078)
Dependency ratio (children and male adults/women)	−0.091* (0.031)	−0.104* (0.034)
Commuting burden	−0.502* (0.097)	−0.570* (0.066)
Constant	0.111 (0.181)	0.539* (0.125)
N (Chi^2)	1235* (52.38)	1189* (96.37)

Notes: Sample restricted to women older than 18 and 60 years or younger. Standard errors in parentheses. *, ** denote significance at the 1 and 5 percent levels, respectively.

and has been accomplished at a very high cost, including environmental costs. In proceeding with these projects, the government has ignored the protests of ordinary citizens, but also the recommendations of technical experts and previous government committees (see Stalin Dayanand [2020] on Mumbai). On the contrary, buses are heavily used but are being neglected (Indorewala 2017). Essentially, individuals and households take the transportation facilities in the city as an exogenous given, which implies that our regression model is justified.

Table 5 presents results from probit regressions. The results for deprived status of the household are on the lines that we described in the decomposition analysis. The positive and statistically significant coefficient on educational attainment confirms our prior expectation that education is positively associated with FLFP (more on this below).³⁵ The dummy for occupation is not statistically significant.³⁶ The negative and statistically significant coefficient on dependency ratio also confirms our hypothesis that care work is negatively associated with FLFP. As we discussed earlier, women face many difficulties in traveling to work within Indian cities. As we can observe from Table 5, in both cities, high commuting burden is negatively associated with the ability of a woman to participate in the labor force.

What can we say about the magnitudes of the impact of care and commuting? Since this is a probit regression, unlike in an OLS regression, we cannot just examine the coefficient of the variable that we are interested

in (care or commuting). We must consider the values of the other variables too. We therefore consider the following case: a woman who has higher secondary or higher education, belongs to a deprived household, is in the informal or self-employed sector, has an average dependency ratio, and lives in an average commuting neighborhood. The marginal effects for dependency ratio (about -0.04) are comparable for the two cities and imply that a unit increase in the dependency ratio is associated with a decrease in the probability of being in the labor force by about 4 percentage points. The marginal effects of commuting are also similar for the cities (about -0.2). Increase in commuting burden by a unit (that is, average distance by a zone) is associated with a decrease in the probability of being in the labor force by about 20 percentage points.

Robustness checks

We have conducted several checks to ensure that the above results are robust.³⁷ Some studies on India have found that FLFPR exhibits a U-shaped relationship with respect to education (Chatterjee, Desai, and Vanneman 2018). To test for this, we estimated a model with years of education and a squared term for years of education. The estimates are presented in Table 6. The results are consistent with a U-shaped relationship in Hyderabad, with a turning point of about seven years. The U-shaped relationship in Mumbai is weak (since the coefficient on years of education is not statistically significant), and the turning point is about five and half years. Note that since the turning points for both Hyderabad and Mumbai are below higher secondary education, the results from this estimation are consistent with those from the earlier estimation (reported in Table 5). The results for the other variables, in particular dependency ratio and commuting burden, continue to hold for the new specification.

We also considered other ways of thinking about the care burden. For example, we controlled for the presence in the household of young children (less than 6 years), more than one adult women (so that the care burden is shared), and older adults (older than 60 years, usually in-laws). Another check we did is to capture all the categories for caste, religion, and class by considering multiple dummies (rather than one binary dummy). We show the results from these estimations in an Appendix (Table A1). As we can observe, the main results on education, care burden, and commuting still go through in all the estimations.³⁸

DISCUSSION AND CONCLUSIONS

Our analysis has established how space and gender influence each other and how social cleavages like caste and class interact with gender. To reiterate our findings: FLFPR varies across cities as well as different

GENDER AND WORK PATTERNS

Table 6 Probit regression for female labor force participation with years of education (Dependent variable: 1 if a woman participates and 0 if not)

	<i>Hyderabad</i>	<i>Mumbai</i>
Belongs to a deprived household	0.278** (0.116)	0.117 (0.078)
Years of education	− 0.134* (0.023)	− 0.026 (0.024)
Years of education squared	0.009* (0.001)	0.002*** (0.001)
Belongs to an informal or self-employed household	0.031 (0.083)	− 0.001 (0.079)
Dependency ratio (children and male adults/women)	− 0.084* (0.031)	− 0.101* (0.034)
Commuting burden	− 0.498* (0.097)	− 0.576* (0.066)
Constant	0.329*** (0.195)	0.571* (0.152)
<i>N</i> (Chi^2)	1235* (87.04)	1189* (97.73)

Notes: Sample restricted to women older than 18 and 60 years or younger. Standard errors in parentheses. *, **, *** denote significance at the 1, 5, and 10 percent levels, respectively.

zones within cities. Neoliberal city zones have higher FLFPR, mainly because they have a higher proportion of women coming from households with privileged class and social backgrounds that tend to have higher FLFPR. These households also provide paid care work to informal women workers. Yet, these zones fare worse in overall development indicators (such as education and poverty) than those neighborhoods that are more mixed along class and social group axes. State policies should address heightened class inequalities within neoliberal zones and target neighborhoods (particularly ghettos) left behind by neoliberal economic policies.

Our econometric investigation shows that three factors are associated with FLFP: commuting ease, care burden in households, and women's education. These findings resonate with women's experiences in cities everywhere, both in developed (Le Barbanchon et al. 2019) and developing countries (see Anand and Tiwari [2006] for India; Quiros, Mehndiratta, and Ochoa [2014] for Argentina).

For commuting, the role of public transport and the emergence of a commuting culture among women may prove crucial. Public transport can add to the objective and subjective understandings of safety and access to work, especially for women who do not own private vehicles.³⁹ The policy conclusion of this analysis is that cities in general, and Indian cities in particular, should improve their systems of public transport for several

reasons (such as environmental, traffic congestion), but also to encourage women's voluntary paid work.

As many analyses of women's work point out, a disproportionate share of care work is borne by women. This can contribute to women opting out of paid work, even when they are qualified and want to work. Reproductive labor inside Indian households has been shaped by market-oriented reforms that have increased the time needed for women to engage in childcare, healthcare, or the procurement of water (water tankers or public water taps that work only during fixed times of the day). There has been a failure to expand state services, especially for informal workers residing in urban slums and squatter settlements. The state should improve the provisioning of these care services that ease the burden of households. Of course, intrahousehold work-sharing norms also need to change, to make women's decisions to perform paid or unpaid work a matter of voluntary rather than a coerced choice.

Educational enrollment and attainment among Indian girls and women have increased considerably in recent years due to state policies.⁴⁰ However, effects of these policies will arrive with a lag, in the future. The state has to play a more proactive role in this regard by ensuring a uniform quality of education. Mumbai has performed much better in college education or above for women (Table 1 and Table 2), and this is likely to be due to the early emphasis on girls' education in Bombay presidency (under British rule) from the 1820s by Christian missionaries (American, Scottish, and British), linguistic societies (Gujarati, Parsi, and Marathi), as well as Hindu Social Reformers (from 1880s onward). Several incentive schemes were also devised to further women's education, albeit confined to strictly defined women's roles (Anagol-McGinn 1994). The difference between the two cities that started in the colonial/British period continues to the present.

The socio-spatial framework adopted in this study helps analyze factors such as commuting, care work, availability of jobs oriented toward women, women's education, and intra- or inter-city space, all of which are closely intertwined in determining FLFPR. In the spirit of Massey (1994), our key contribution is to show that the work patterns of women in cities reflect both socioeconomic considerations as well as geography.

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- ¹ See Mike Davis (2005). UN (2015) presents trends and projections, according to which, six of the world's ten largest cities will be in the Global South.
- ² Our use of the term "gender" is consistent with scholarship in feminist economics and feminist geography. Gender is relational, is socially constructed, exhibits spatial and temporal variations, and includes among other things, social norms and roles. The issues that we discuss in this article (such as care, commuting, and labor force participation) are not merely concerns of women, but relational in nature and embedded in a societal (urban) and household (patriarchal) context, which manifests in different positive and negative outcomes for women and men. Therefore, we use "gender" (rather than "women") in the title of the article.
- ³ We want to state categorically that we are not privileging paid employment (labor force participation) over not participating in the labor force for women. However, FLFPR is a particular manifestation of gendered city spaces, and we are simply using it as a reflection of how gender and space mutually determine each other.
- ⁴ Throughout the article, we use "labor force" to refer to *paid* labor force. We distinguish between paid and unpaid care work and include the former in the labor force, while the latter is typically performed in the household and is not included in the official definition of labor force. When we use the expression care work/care burden in the article, we typically refer to the latter unless specified as paid care work. In general, richer data are available for cities in the developed world. For the US, data on work and labor force for cities is available from the Bureau of Labor Statistics: <https://www.bls.gov/sae/>.
- ⁵ *Nirbhaya* ("fearless") is the name given by Indian media to the sexual assault and murder of Jyoti Singh, a physiotherapy student in New Delhi in 2012. This case garnered widespread attention and resulted in protests throughout India. Six people were convicted. (<https://timesofindia.indiatimes.com/topic/nirbhaya-gang-rape>).
- ⁶ Other common databases are: National Family Health Survey (NFHS), India Human Development Survey (IHDS), and the decennial Census. The census contains some information on intra-city spatial units and presents transport patterns at the district level. However, it allows limited scope for individual or household-level analysis and restricts analysis on other dimensions (for example, caste).
- ⁷ A time-use survey was conducted for India in 2019. Before this, only one time-use survey covering all regions of India had existed (Hirway and Jose 2011).
- ⁸ Rahul Lahoti and Hema Swaminathan (2016) test for a U-shaped relationship between economic development and FLFPR.
- ⁹ For reasons discussed below, we advocate caution in interpreting our findings as causal.
- ¹⁰ For Maharashtra, see: <https://www.maharashtra.gov.in/1128/Districts>. Mumbai UA is spread across Mumbai and Mumbai Suburban districts and parts of Thane district. At the time of our survey, Hyderabad UA was spread across Hyderabad district and parts of Rangareddy and Medak districts. Telangana was reorganized recently, see: <https://www.telangana.gov.in/about/districts>.
- ¹¹ EBs were selected using the popular sampling methodology of PPS (Probability Proportional to Size), which ensures that larger EBs are more likely to be selected into the sample. In Mumbai, we were able to administer the survey to 980 households. The data from the survey, with all relevant information and details, will be made available online soon.
- ¹² The survey was designed carefully, and a pilot was administered before it was finalized. Many questions required respondents to pick a response from a set of possibilities (for

- example, caste – ST, SC, OBC, or Others), but there were also open-ended questions (for example, Occupation).
- ¹³ The literature on caste is too large to summarize here. See Dipankar Gupta (2000), Nicholas B. Dirks (2001), R. S. Rawat and K. Satyanarayana (2016), and the references therein.
 - ¹⁴ It is important to note that “Others” have a much higher proportion in Mumbai than in Hyderabad. This has a historical origin. In early modern Maharashtra, a local leader, Shivaji, mobilized a multi-caste alliance and gave them a new common identity – Marathas, who are all included in Hindu Others. Maharashtra and Mumbai, therefore, have a higher proportion of Others than most states in India.
 - ¹⁵ See the full list at: <http://www.icssrdataservice.in/datarepository/index.php/catalog/89/download/1138>.
 - ¹⁶ The Indian government did not make the latest consumption expenditure survey (2017–18) publicly available. Given this, we are unable to compare estimates from our survey with Hyderabad and Mumbai estimates from publicly available and nationally representative secondary data. The latest Periodic Labor Force Survey (PLFS; 2018–19) only presents city-level estimates of Labor Force Participation for those ages 15 years or higher: 23 percent (Greater Hyderabad Municipal Corporation [GHMC]) and 34 percent (Greater Mumbai Municipal Corporation [GMC]; PLFS 2019, Table 48). These estimates compare well with those from our survey: 28 percent (Hyderabad) and 32 percent (Mumbai). As we discussed above, our survey region of Hyderabad covers a large part (and not the entire area) of GHMC. The slight departures should be seen in light of this.
 - ¹⁷ See Manzoor Alam (1973) for a history of Hyderabad.
 - ¹⁸ According to the latest (2011) Census, the populations of these zones (in millions) are approximately: Old-Walled (1.60), Nizam (1.01), British Resident (1.03), and Neoliberal (0.31).
 - ¹⁹ For estimates of percentages who use these modes from the latest (2011) Census, see: https://www.censusindia.gov.in/2011census/hlo/Data_sheet/India/Transportation.pdf.
 - ²⁰ As mentioned above, India uses an NCO (Government of India 2004) that has one-digit, two-digit, and three-digit codes for various occupations. According to the 2018–19 PLFS, occupations with one-digit codes – 2 (Professionals) and 3 (Technicians and Associate Professionals) – are the top two in urban India in terms of the relative prevalence of women (difference between the proportions of men and women involved). Hence, we term these as “women-oriented” jobs at the present (without attributing any permanence or sanctity to this). Some examples of such jobs are: computer professionals, college and university teachers, nurses, and school teachers. 4 (Clerks) and 9 (Elementary Occupations: low paid/working class) also have relatively high prevalence of women, but less so than 2 and 3. Needless to say, all jobs can be performed by both women and men.
 - ²¹ In our previous work, we found that more households living in more mixed neighborhoods are less likely to be poor, and adults living in such neighborhoods are more educated.
 - ²² For a history, see Sharada Dwivedi and Rahul Mehrotra (1995) and Mariam Dossal (2010). The city’s name was changed from Bombay to Mumbai in 1995.
 - ²³ According to the latest Census, the populations of these zones (in millions) are approximately: British Neoliberal (1.22), Old Industrial 1 (1.87), Western Neoliberal (3.34), Northern Neoliberal (3.09), and Old Industrial 2 (2.93).
 - ²⁴ On Bandra-Kurla-Complex (BKC), a major business center in this area, see Clara Lewis (2018).

- ²⁵ “Old Industrial City 1” has seen the emergence of malls and elite enclaves after the collapse of the textile industry by the 1980s. However, it still carries traces of the old and the *relative* imprint of neoliberalism is less compared to the other zones where we have used the adjective “Neoliberal.”
- ²⁶ Figures from 2011 Census: censusindia.gov.in, On_Slums-2011Final.
- ²⁷ See, for example, *Ladies Special* (<https://www.imdb.com/title/tt1719849/>), a TV program.
- ²⁸ See, for example, <https://scroll.in/video/804816/the-unique-rhythms-of-life-on-mumbai-s-local-trains>; <https://www.cntraveller.in/story/mumbai-local-train-7-things-i-miss-andheri/>.
- ²⁹ Consider a population that is divided into $G (> 1)$ subgroups. Let the size, share and $FLFPR$ of sub-group g be denoted: N_g , s_g and $FLFPR_g$, respectively. Total $FLFPR = \sum_{g=1}^G s_g FLFPR_g$ and contribution of g is: $\frac{s_g FLFPR_g}{FLFPR}$.
- ³⁰ These could also be described as “demand-side,” that is, related to availability of jobs and amenities, most importantly transport infrastructure. We prefer the term “spatial” because what is important is that these jobs and amenities are available in the neighborhood/spatial unit of residence.
- ³¹ Note that this dummy depends largely upon the occupation of a male member (for example, father or husband) and therefore whether a woman participates in the labor force or not has no bearing on it (that is, no issue of reverse-causality).
- ³² We considered individuals younger than 18 years as children. For an overview of care work in developing countries and its impact on economic development, see Nancy Folbre (2018).
- ³³ Good data that can be used as a proxy for *access* to transportation to various parts of the city, that can be used in a regression like ours is unfortunately unavailable. We are therefore focusing on *distance*. We do believe that distance does a good job of capturing the commuting burden.
- ³⁴ In the 2018–19 PLFS, more than 90 percent of never-married women in urban India are either unmarried daughters or unmarried granddaughters of the head of the household.
- ³⁵ Jayan Jose Thomas (2010) uses various rounds of the NSS data on Employment and Unemployment Situation to examine jobs created during the period 1993–2005 at the all-India and sectoral (rural and urban) levels. The finding that is pertinent for us is that job-creation has occurred disproportionately for less-educated (primary or middle school) individuals and urban educated women have not fared well during this period. This is not inconsistent with what we have found because we are focusing on two specific large cities (and not the entire urban sector, which consists of small towns and various kinds of cities – smaller and larger), at a point in time and almost one and half decades after 2004–05 (when Thomas’s study ends). Moreover, unlike Thomas (2010), since our analysis is regression based, we are able to control for other factors while examining the impact of education.
- ³⁶ The result on education is consistent with some studies and in conjunction with the result on occupation casts doubts on the idea that increased incomes are resulting in withdrawals (Kannan and Raveendran 2012).
- ³⁷ In the interests of space, we do not report all the estimations/robustness checks that we performed. They are available on request.
- ³⁸ There are cost-of-living differences between the two cities. In particular, Mumbai is considered to be an expensive city compared to Hyderabad. We have conducted the regression analysis separately for the two cities because it is realistic to assume that FLFP does not follow the same model in the two cities. However, we pooled the

data from the two cities and ran one regression with a Mumbai city dummy (which is equivalent to including a variable that captures cost of living in either city). We found that the Mumbai city dummy has the right sign (positive) and is statistically significant.

- ³⁹ In Indian cities, some measures for the safety and comfort of women (for example, through reserved seats) are in place. In our opinion, these are inadequate and need to be expanded.
- ⁴⁰ See the studies that we cited in the Introduction, for example, C. Rangarajan, P. Iyer, and Seema (2011). This increased enrollment is reflected in the surveys on Employment and Unemployment Situation conducted over various years. The previous Indian governments adopted measures to improve enrollment in education notably Sarva Siksha Abhiyan (Education for All) and Right to Education (RTE) Act.

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GENDER AND WORK PATTERNS

APPENDIX

Table A1 Probit regression for female labor force participation (dependent variable: 1 if a woman participates and 0 if not)

	<i>Hyderabad</i>	<i>Hyderabad</i>	<i>Mumbai</i>	<i>Mumbai</i>
Dalit	0.388*	0.435*	− 0.018	− 0.018
	(0.134)	(0.136)	(0.109)	(0.109)
OBC	0.402*	0.399*	0.018	0.020
	(0.093)	(0.095)	(0.122)	(0.123)
Hindu	0.060	0.084	− 0.176***	− 0.176***
	(0.096)	(0.097)	(0.099)	(0.100)
Non-Hindu and non-Muslim	0.276	0.240	0.212	0.232
	(0.194)	(0.198)	(0.168)	(0.169)
Elite	0.264***	0.351**	0.516*	0.541*
	(0.162)	(0.165)	(0.172)	(0.173)
Professional	− 0.120	− 0.069	− 0.067	− 0.064
	(0.141)	(0.143)	(0.140)	(0.140)
Formal worker	0.017	0.043	0.057	0.079
	(0.098)	(0.100)	(0.096)	(0.096)
Informal owner	0.148	0.133	− 0.176	− 0.157
	(0.137)	(0.138)	(0.158)	(0.159)
Retired or others	− 0.844*	− 0.810*	− 0.626*	− 0.543*
	(0.237)	(0.238)	(0.246)	(0.248)
Years of education	− 0.134*	− 0.142*	− 0.032	− 0.035
	(0.024)	(0.024)	(0.024)	(0.024)
Years of education squared	0.009*	0.010*	0.003***	0.003**
	(0.001)	(0.001)	(0.001)	(0.001)
Dependency ratio (children and male adults/women)	− 0.075*		− 0.108*	
	(0.032)		(0.034)	
Young children at home		− 0.540*		− 0.225**
		(0.092)		(0.099)
Older adults at home		− 0.133		0.048
		(0.094)		(0.107)
Only adult woman at home	0.138		− 0.167**	
	(0.088)		(0.084)	
Communting burden	− 0.417*	− 0.411*	− 0.545*	− 0.533*
	(0.102)	(0.104)	(0.068)	(0.069)

(Continued)

Table A1 Continued.

	<i>Hyderabad</i>	<i>Hyderabad</i>	<i>Mumbai</i>	<i>Mumbai</i>
Constant	0.189 (0.196)	0.130 (0.192)	0.726* (0.172)	0.595* (0.160)
N (χ^2)	1235* (137.140)	1235* (176.480)	1189* (124.930)	1189* (124.950)

Notes: Sample restricted to women older than 18 and 60 years or younger. Standard errors in parentheses. *, **, *** denote significance at the 1, 5, and 10 percent levels, respectively.